

Confidence Intervals (Part III)

STAT 24510-30040

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6 Variable Stabilizing Transformation Method

Variance-stabilizing transformation (VST) is a data-transformation technique in data analysis practice.

The method is attributed to British statistician F. J. Anscombe.

The VST method provides another way to find asymptotically pivotal quantities and to construct asymptotic confidence intervals.

A key step in the derivation of variance-stabilization transformation is the delta method or δ -method.

The delta method itself is a very useful procedure in statistics, frequently used to linearize a non-linear problem.

Motivation of VST

Consider an unbiased estimator T of a parameter θ of interest, with $\mathbb{E}(T) = \theta$, $Var(T) = v(\theta)$.

For example, the parameter of interest θ can be the population mean and $T = \bar{X}$ the sample average (depending on n).

When sample observations $X_i, i = 1, \dots, n$ are considered i.i.d. of mean θ , $\mathbb{E}(T) = \mathbb{E}(\bar{X}) = \theta$, T is unbiased.

If X_i 's are of $N(\theta, \sigma^2)$, then $Var(T) = \frac{\sigma^2}{n}$ does not vary with the mean θ , whether σ^2 is known, or estimated from data.

But that is as good as it gets. For other distributions, the $Var(T)$ will vary with the population mean θ .

For example, if X_i 's are i.i.d. $\sim Poisson(\theta)$, then $Var(\bar{X}) = \frac{\theta}{n}$. The variance is very much a function of θ .

When variance of $\hat{\theta}$ (or a function $\hat{\theta}$ which is of interest) depends on the unknown parameter θ , it can be inconvenient in certain inference, especially when estimating the parameter θ itself is a main objective, having variance depending on θ making us uncertain about the uncertainty of θ .

Variance-stabilizing method looks for a transformation $g(T)$ of the data thus the statistic T , with the objective that the variance of $g(T)$ is a constant not depending on θ , possibly approximately. The common assumption is that g is a differentiable function.

The derivation of variance-stabilizing transformation utilizes the δ method, an approximation method base on the Taylor polynomial approximation of smooth functions.

Review Taylor's polynomial approximation

If real function $g(x) : \mathbb{R} \rightarrow \mathbb{R}$ is k th differentiable at $a \in \mathbb{R}$, then for x near a , $g(x)$ can be approximated by the k -th Taylor polynomial $P_k(x)$,

$$g(x) \approx P_k(x)$$

with

$$P_k(x) = g(a) + g'(a)(x-a) + \frac{g''(a)}{2!}(x-a)^2 + \dots + \frac{g^{(k)}(a)}{k!}(x-a)^k$$

The rate of the approximation can be stated as

$$\lim_{x \rightarrow a} \frac{g(x) - P_k(x)}{(x-a)^k} = 0$$

Therefore approximation is best used for x near a .

For example, when $g''(x)$ exists, the function can be approximated by a quadratic function,

$$g(x) \approx P_2(x) = g(a) + g'(a)(x-a) + \frac{g''(a)}{2}(x-a)^2$$

with the rate of the approximation

$$\lim_{x \rightarrow a} \frac{g(x) - P_2(x)}{(x-a)^2} = 0$$

The approximation can be expressed using the little "o" notation.

$$g(x) = g(a) + g'(a)(x-a) + \frac{g''(a)}{2}(x-a)^2 + o((x-a)^2)$$

meaning the remainder term $o((x-a)^2)$ converges to 0 faster than $(x-a)^2$ as $x \rightarrow a$.

The notation "o" is called "little o". A function $h(t)$ is $o(t)$ if and only if

$$\lim_{t \rightarrow 0} \frac{h(t)}{t} = 0$$

In particular, when g is differentiable, the function can be approximated by the tangent line of the function $g(\cdot)$ at a ,

$$g(x) \approx g(a) + g'(a)(x-a) \quad (1)$$

in the sense that

$$g(x) = g(a) + g'(a)(x-a) + o(x-a)$$

This linear approximation is used in the following derivation.

We will consider the case that function g is differentiable, and the statistic T has finite variance.

We look for suitable functions g such that the variance of $g(T)$ will be approximately constant, not depending on θ .

The delta method

By the first order Talyor polynomial approximation (1), for values of T near θ , $g(T)$ has the approximation

$$g(T) \approx g(\theta) + g'(\theta)(T - \theta) \quad (2)$$

which represents random variable $g(T)$ as a linear function of random variable T .

The method is known as the δ method, or propagation of error in some applications, often used to linearize a nonlinear problem.

Take expectation on both side of (2). Note that $g(\theta), g'(\theta)$ are fixed, non-random quantities. We obtain

$$\mathbb{E}[g(T)] \approx g(\theta) + g'(\theta) \mathbb{E}(T - \theta)$$

We consider the case of unbiased T with $\mathbb{E}(T) = \theta$, or at least approximately, $\mathbb{E}(T) \approx \theta$. Then $T - \theta$ should be small (e.g., when sample size n is large), $E(T - \theta) \approx 0$.

Then the first order approximation of the Taylor expansion implies

$$\mathbb{E}[g(T)] \approx g(\theta)$$

which leads to its variance approximation

$$Var[g(T)] = \mathbb{E}((g(T) - \mathbb{E}[g(T)])^2) \approx \mathbb{E}([g(T) - g(\theta)]^2)$$

Rewrite (2) as

$$g(T) - g(\theta) \approx g'(\theta)(T - \theta)$$

Square both sides then take expectation, the approximation gives

$$Var[g(T)] \approx \mathbb{E}([g(T) - g(\theta)]^2) \approx g'(\theta)^2 \mathbb{E}[(T - \theta)^2] = g'(\theta)^2 Var(T) = g'(\theta)^2 v(\theta)$$

where $v(\theta) = Var(T)$ typically is a function of θ .

Derivation of the variance stabilizing transformation

Now we have an explicit (approximate) expression of the variance of $g(T)$ in terms of $Var(T)$ and $g'(\cdot)$,

$$Var[g(T)] \approx g'(\theta)^2 v(\theta) \quad (3)$$

from which we will choose a variance stabilizing transformation function $g(\cdot)$.

In order to achieve constant-variance after the transformation, we need

$$Var[g(T)] \approx g'(\theta)^2 v(\theta) = constant$$

meaning not depending on unknown parameters. We may choose a function g such that

$$g'(\theta) = \frac{1}{\sqrt{v(\theta)}} \quad (4)$$

(or its constant multiple). This transformation will make

$$Var(g(T)) \approx 1$$

The variance does not depend on θ , just as desired. We solve for a $g(\cdot)$ satisfying (4) by integration,

$$g(\theta) = \int \frac{1}{\sqrt{v(\theta)}} d\theta$$

More generally, if we just need constant variance rather than variance = 1, we may choose any g up to a constant multiple, as long as

$$g'(\theta) \propto \frac{1}{\sqrt{v(\theta)}}$$

which means left hand side is proportional to the right hand side (by a constant multiplier).

The resulting statistic $g(T)$ will be approximately unbiased and with constant variance,

$$\mathbb{E}[g(T)] \approx g(\theta), \quad Var(g(T)) \approx constant \quad (\text{does not vary with } \theta)$$

As in majority applications, if the original estimator T is of normal distribution, exactly or asymptotically,

$$T \sim N(\theta, v(\theta))$$

Linear transformation of normal random variables is also of normal distribution. Since the δ -method by Taylor linear approximation (2) means that random variable $g(T)$ is approximately a linear transformation of random variable T , thus $g(T)$ also approximately of normal distribution,

$$g(T) \sim N(g(\theta), g'(\theta)^2 v(\theta)) \quad (\text{approximately})$$

Our choice of $g'(\theta)^2 v(\theta) = constant$ means

$$g(T) \sim N(g(\theta), constant) \quad (\text{approximately})$$

Example: Poisson VST and C.I.

Data: $X_1, \dots, X_n$.

Model: X_i 's are *i.i.d.* $\sim Poisson(\lambda)$.

Goal: Derive a variance stabilizing transformation, then construct a confidence interval for λ based on the VST.

From the Central Limit Theorem, the MLE $\hat{\lambda} = \bar{X}$ has asymptotic distribution

$$\frac{\hat{\lambda} - \lambda}{\sqrt{\lambda/n}} \sim N(0, 1) \quad (\text{approximately})$$

which means, for large n ,

$$\hat{\lambda} - \lambda \sim N\left(0, \frac{\lambda}{n}\right) \quad \text{or} \quad \sqrt{n}(\hat{\lambda} - \lambda) \sim N(0, \lambda) \quad (5)$$

The goal is to find a function $g(\cdot)$, such that

$$\sqrt{n}(g(\hat{\lambda}) - g(\lambda)) \sim N(0, 1)$$

asymptotically approximately. By the relation (2) using the delta method,

$$g(\hat{\lambda}) - g(\lambda) \approx g'(\lambda)(\hat{\lambda} - \lambda)$$

So approximately, the random variable $g(\hat{\lambda})$ is a linear transformation of the random variable $\hat{\lambda}$.

Since $\hat{\lambda}$ has the asymptotic normal distribution in (5), $g(\hat{\lambda})$ as a linear transformation of $\hat{\lambda}$ is also asymptotically normal, with variance derived as in (3).

$$g(\hat{\lambda}) - g(\lambda) \sim N(0, g'(\lambda)^2 Var(\hat{\lambda}))$$

Equivalently,

$$\sqrt{n}(g(\hat{\lambda}) - g(\lambda)) \sim N(0, g'(\lambda)^2 n Var(\hat{\lambda}))$$

Recall that $Var(\hat{\lambda}) = \lambda/n$, thus we have

$$\sqrt{n}(g(\hat{\lambda}) - g(\lambda)) \sim N(0, g'(\lambda)^2 \lambda)$$

Recall that our goal is to find g such that

$$\sqrt{n} \left(g(\hat{\lambda}) - g(\lambda) \right) \sim N(0, 1)$$

Thus, the function $g(\cdot)$ should satisfy

$$g'(\lambda)^2 \lambda = 1$$

If we choose $g'(\cdot) > 0$,

$$g'(\lambda)^2 \lambda = 1 \iff g'(\cdot) = \frac{1}{\sqrt{\lambda}} \iff g(\lambda) = \int g'(\cdot) d\lambda = 2\sqrt{\lambda} + c$$

We can choose

$$g(\lambda) = 2\sqrt{\lambda}$$

Now we have achieved

$$\sqrt{n} \left(g(\hat{\lambda}) - g(\lambda) \right) = 2\sqrt{n} \left(\sqrt{\hat{\lambda}} - \sqrt{\lambda} \right) \sim N(0, 1)$$

for large n . Therefore,

$$\mathbb{P} \left(z_{\alpha/2} \leq 2\sqrt{n} \left(\sqrt{\hat{\lambda}} - \sqrt{\lambda} \right) \leq z_{1-\alpha/2} \right) \approx 1 - \alpha$$

Consequently, we obtain an $(1 - \alpha)100\%$ asymptotic confidence interval by the VST method for $\sqrt{\lambda}$ as

$$\left[\sqrt{\hat{\lambda}} - \frac{z_{1-\alpha/2}}{2\sqrt{n}}, \sqrt{\hat{\lambda}} + \frac{z_{1-\alpha/2}}{2\sqrt{n}} \right]$$

Note that the left endpoint of the VST confidence interval could be < 0 . Since $\lambda \in (0, \infty)$, in practice, the left endpoint should be truncated to 0 whenever it falls < 0 . So the VST asymptotic confidence interval for $\sqrt{\lambda}$ should be truncated as

$$\left[\max \left\{ 0, \sqrt{\hat{\lambda}} - \frac{z_{1-\alpha/2}}{2\sqrt{n}} \right\}, \sqrt{\hat{\lambda}} + \frac{z_{1-\alpha/2}}{2\sqrt{n}} \right]$$

From this confidence interval for $\sqrt{\lambda}$, we can square the values of the endpoints to obtain a $(1 - \alpha)100\%$ asymptotic confidence interval for λ , in the form

$$\left[\left(\max \left\{ 0, \sqrt{\hat{\lambda}} - \frac{z_{1-\alpha/2}}{2\sqrt{n}} \right\} \right)^2, \left(\sqrt{\hat{\lambda}} + \frac{z_{1-\alpha/2}}{2\sqrt{n}} \right)^2 \right]$$

Remarks

- In the above, we applied variance stabilization transformation for Poisson parameter as

$$g(\theta) = 2\sqrt{\theta} \implies \text{Var} \left(g(\hat{\theta}) \right) \approx 1.$$

In practice, it is more common to use

$$g(\theta) = \sqrt{\theta} \implies \text{Var} \left(g(\hat{\theta}) \right) \approx \frac{1}{4}.$$

- As we derived in class, Poisson is a type of limit of binomial distribution. Recall that a binomial random variable is already a type of sum. Thus for moderate to large λ , the CLT implies that the MLE $\hat{\lambda}$ for $\text{Poisson}(\lambda)$ is approximately normal (when $n = 1$),

$$\hat{\lambda} = X \sim N(\lambda, \lambda) \quad \text{asymptotically.}$$

Consequently, using variance stabilizing transformation $g(\lambda) = \sqrt{\lambda}$,

$$X \sim \text{Poisson}(\lambda) \implies \sqrt{X} \sim N \left(\sqrt{\lambda}, \frac{1}{4} \right)$$

Comments on variance stabilization

- Variance stabilization is useful in comparison distributions in the same family with different parameter values.
- Variance stabilization provides yet another method to conduct hypothesis testing and to obtain confidence intervals.
- Variance stabilization induced confidence interval may not provide the best coverage probability. The induced confidence intervals sometimes are too wide, due to multiple steps of approximations.
- Second order approximation

$$g(T) \approx g(\theta) + g'(\theta)(T - \theta) + \frac{g''(\theta)}{2}(T - \theta)^2 \quad (6)$$

yields a better approximation of the mean

$$\mathbb{E}[g(T)] \approx g(\theta) + \frac{\sigma_{\hat{\theta}}^2}{2} g''(\theta), \quad \sigma_{\hat{\theta}}^2 = \mathbb{E}(\hat{\theta} - \theta)^2$$

- There are refinements of the variance stabilization transformations.
For example, $\sqrt{Y + a}$ has been proposed for Poisson, suggested a values include $1/2$ and $3/8$.
- In linear regression models, variance stabilization can be useful in maintaining constant variance across covariate space.
- Though there are certain similarities in the form of transformation, the Jeffrey prior transformation in Bayesian statistics stems from a very different idea of invariance, and it is for parameters (and their non-informative priors), not for data sample statistics as in the variance stabilization discussed here.

Relevant sections in Rice: 4.6, 9.7.